

Ex.No:1: Simulation of FDMA in MATLAB Date.

Objective 1:

To simulate an FDMA system by allocating individual frequency channels to multiple users and determine the channel bandwidth assigned to each user.

Program:

```
clc; clear; close all;
fs = 1000;
t = 0:1/fs:1;
fc = [100 300 500];
m = sin(2*pi*10*t);
subplot(2,1,1)
hold on
for i = 1:3
plot(t, m.*cos(2*pi*fc(i)*t))
end
grid on
title('FDMA Channel Allocation')
xlabel('Time (s)')
ylabel('Amplitude')
legend('User 1','User 2','User 3')
subplot(2,1,2)
bar(1:3, 200*ones(1,3))
grid on
title('Channel Bandwidth per User')
xlabel('Users')
ylabel('Bandwidth (Hz)')
```

Output:

Objective 2:

To calculate the data rate achieved by each user in the FDMA system based on the allocated channel bandwidth.

Program:

```
clc; clear; close all;
BW = 200; % Channel Bandwidth (kHz)
Users = 3;
M = 2; % BPSK Modulation
```

```

DataRate = BW*log2(M); % Data Rate (kbps)
Rate = DataRate*ones(1,Users);
disp(' User Data Rate (kbps)')
disp([(1:Users)' Rate'])
figure
bar(1:Users,Rate)
xlabel('Users')
ylabel('Data Rate (kbps)')
title('FDMA Data Rate for Each User')
grid on

```

Output:

Objective 3:

To evaluate the communication quality of the FDMA system by calculating the Signal-to Noise Ratio (SNR) for different users.

Program:

```

clc; clear; close all;
Ps = 1; % Signal Power (W)
Pn = [0.1 0.05 0.01]; % Noise Power (W)
Users = 1:3;
SNR = 10*log10(Ps./Pn); % SNR in dB
disp('User Noise Power(W) SNR(dB)')
disp([Users' Pn' SNR'])
figure
bar(Users,SNR)
xlabel('Users')
ylabel('SNR (dB)')
title('Signal-to-Noise Ratio of FDMA Users')
grid on

```

Output: